

Introduction to Duration Models

Introduction aux modèles de durée

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Agenda

Introduction

Organization

Overview

Lectures' plan

The law of a duration

Censoring and truncation

Likelihood inference

Nonparametric inference under random right-censoring

Duration analysis in presence of covariates

Module organization

- ▶ There are 4 lectures, 2 TD and 1 TP
- ▶ Evaluation
 - ▶ Quiz during the TD/TP (possibly at the end of the lectures)
 - ▶ Final exam, no documents (*formulaire* allowed)

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Some references

- ▶ J.P. KLEIN, M.L. MOESCHBERGER, *Survival Analysis : Techniques for Censored and Truncated Data (2nd edition)*, Statistics for Biology and Health, Springer, 2003.
- ▶ J.P. KLEIN, H.C. van HOUWELINGEN, J.G. IBRAHIM, T.H. SCHEIKE: *Handbook of Survival Analysis*, CRC Press, 2014.
- ▶ D.G. KLEINBAUM, J.P. KLEIN: *Survival Analysis. A Self-Learning Text*, Springer, 2005.
- ▶ Lecture Notes in French available online
 - ▶ P. SAINT PIERRE, *Intro à l'analyse des durées de survie*.
https://perso.math.univ-toulouse.fr/psaintpi/files/2021/04/Cours_Survie_1.pdf
 - ▶ C. HUBER, *Modèles pour des durées de survie*.
docplayer.fr/423140-Modeles-pour-des-durees-de-survie.html

Avant-propos

- ▶ In this course we study the ***time-to-event***
- ▶ The event could be birth, failure, default, disease onset, death, divorce, employment, get connected on social networks *etc.*
- ▶ Many potential applications in engineering, finance, economics, biology and medicine, social networks *etc.*
- ▶ Depending on the domain, the statistical methodologies used to study the time-to-event are called *duration models*, *survival analysis*, *reliability analysis*

Why a specific statistical area?

- ▶ Like usually, the statistician' aim is to model the law of the observations, that is of the realizations of duration random variable. The interest could be
 - ▶ the marginal law (no covariates)
 - ▶ or the conditional law given some covariates (regression context)
- ▶ **Remark:** the time series framework will not be considered
→ too complicated and not really useful for the applications
- ▶ The same types of approaches will be investigated
 - ▶ parametric
 - ▶ semiparametric
 - ▶ nonparametric

- ▶ However, there are some specific characteristics of the law of the observations that are of interest in the time-to-event analysis
- ▶ In particular, in general, one is not only interested in some characteristics of the law, or of the conditional law. One would rather look for a full description of the (marginal or conditional) law.
- ▶ Moreover, the realizations of the duration variables are not always available – specific reasons of incompleteness are usually considered in time-to-event analysis, such as
 - ▶ censoring (right, left, interval, type I, type II *etc.*)
 - ▶ truncation

Big data and the bias

- ▶ Modern applications provide large/huge/massive samples
- ▶ Basically, this implies estimates with (very) small/tiny variance
- ▶ In many situations, a non-negligible part of the data is not accurate/complete
 - ▶ missing observations
 - ▶ observations with errors, segmented,...
 - ▶ censored, truncated,...
 - ▶ ...
- ▶ 'Incomplete' data can lead to bias, whatever the sample size is!

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Lecture 1: objectives

- ▶ State the general interests and objectives one has in time-to-event analysis, with or without predictors (regressors)
- ▶ Present probabilistic tools used in time-to-event analysis
 1. alternative ways, more adapted ways, to define probability measures and thus statistical models
 2. hazard functions, mean residual life, quantiles, relationships
 3. common probability distributions in time-to-event analysis
- ▶ Present specific features of this type of analysis
 1. some observation units are not entirely observed due to specific incompleteness mechanisms (censoring, truncation);
 2. incomplete observations carry information and thus should be treated in a correct way to avoid biases

Lecture 2: objectives

- ▶ Introduce the typical incompleteness mechanisms
 1. censoring and truncation
 2. focus on random right-censoring
 3. highlight the possible bias in inference if censoring is not taken into account properly
- ▶ Recall the likelihood principle, present the likelihood principle under censoring or/and truncation
 1. building the likelihood under censoring/truncation
 2. parametric likelihood examples under censoring
- ▶ Nonparametric inference without covariates
 1. Survival function and cumulative hazard function nonparametric estimators
 2. The random right-censoring case: Kaplan-Meier and Nelson-Aalen estimators

Lecture 3: objectives

- ▶ Asymptotic Theory for Kaplan-Meier (KM) estimator
 1. From empirical distribution function to KM estimator
 2. The law of large number and the CLT
 3. Interpreting the KM integrals
- ▶ Testing equality of the survival functions of different groups
 1. Log-rank test
- ▶ Introduction to the predictive analysis of durations
 1. Parametric regression duration models

Lecture 4: objectives

- ▶ Further explore the regression context in duration models
 1. Proportional hazard models, Cox semiparametric model
 2. Accelerated Failure Models
 3. Conditional Kaplan-Meier estimators
- ▶ Discussions and conclusions
 1. Take away ideas
 2. Some current research topics

Can I use AI for these lectures?

- ▶ Yes, you can! Perhaps you should!
- ▶ BUT...

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Introduction

The law of a duration

- Characterizing the law of a duration

- Classifying the duration laws

- Stochastic ordering

- Common laws in duration models

Censoring and truncation

Likelihood inference

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Ways to characterize a probability distribution (1/2)

- ▶ Assume that we study a random variable $Y \in \mathbb{R}_+ = [0, \infty)$ that represent a time-to-event.
- ▶ Distribution function: $F_Y : \mathbb{R}_+ \mapsto [0, 1]$ defined by

$$F_Y(y) = \mathbb{P}(Y \leq y), \quad y \geq 0$$

- ▶ **Survival function:** $S_Y : \mathbb{R}_+ \mapsto [0, 1]$ defined by

$$S_Y(y) = \mathbb{P}(Y > y) = 1 - F_Y(y), \quad y \geq 0$$

- ▶ Some authors denote the survival function by $\bar{F}_Y$, and called it *survivor* or *reliability* function.
- ▶ **Notation:** $S_Y(y-) = \mathbb{P}(Y \geq y), y \geq 0$

Ways to characterize a probability distribution (2/2)

- ▶ We say that Y admits a density (with respect to the Lebesgue measure on the real line) if there exists a measurable function $f_Y : \mathbb{R}_+ \mapsto \mathbb{R}_+$ such that

$$F_Y(y) = \int_0^y f_Y(t)dt, \quad y \geq 0 \quad \left(\text{also } S_Y(y) = \int_y^\infty f_Y(t)dt, \quad y \geq 0 \right)$$

In this case, we also say that Y is absolutely continuous, or simply *continuous*

- ▶ If $F_Y(\cdot)$ is differentiable at t , then

$$f_Y(y) = F'_Y(y) = -S'_Y(y)$$

- ▶ In the following we also use the simplified notation f , F and S (or $\bar{F}$) instead of f_Y , F_Y and S_Y (or $\bar{F}_Y$)
- ▶ Other ways to characterize the law of Y : characteristic function, moment generating function, *etc.*

Exercise: Moments Using the Survival Function

- ▶ Let Y be a nonnegative random variable with survival function S .
- ▶ **Exercise:** Show that, for any $\alpha > 0$,

$$\mathbb{E}(Y^\alpha) = \alpha \int_0^\infty y^{\alpha-1} S(y) dy,$$

in the sense that if one side converges so does the other.¹

- ▶ Deduce

$$\mathbb{E}(Y) = \int_0^\infty S(y) dy.$$

- ▶ **Exercise.** Propose an alternative, direct proof for the relationship

$$\mathbb{E}(Y) = \int_0^\infty S(y) dy$$

using Fubini's Theorem.

¹ See Feller (1966), *An Introduction to Probability Theory and Its Applications*, vol. 2, Lemma 1, p. 150.  19/122

Hazard functions : discrete random variables (1/2)

- ▶ Let $Y \in \{y_1, y_2, \dots\}$ with $0 \leq y_1 < y_2 < \dots$
- ▶ Let $p_k = \mathbb{P}(Y = y_k) > 0$, $k \geq 1$
- ▶ The *hazard function* (also called *hazard rate*, or *failure rate*) is defined as

$$\lambda(y_k) = \frac{\mathbb{P}(Y = y_k)}{\mathbb{P}(Y \geq y_k)} = \frac{p_k}{\sum_{j \geq k} p_j} = \frac{p_k}{S(y_{k-1})} = \frac{p_k}{S(y_k-)}, \quad k \geq 1.$$

- ▶ We could also write the hazard function as a conditional probability

$$\lambda(y_k) = \mathbb{P}(Y = y_k \mid Y \geq y_k)$$

- ▶ The hazard rate is thus the probability that the event occurs at time y_k given that it did not occur previously

Hazard functions : discrete random variables (2/2)

- **Exercise:** show that for each $k \geq 1$,

$$\lambda(y_k) = 1 - \frac{S(y_k)}{S(y_{k-1})} = 1 - \frac{S(y_k)}{S(y_{k-})}, \quad S(y_k) = \prod_{j=1}^k \{1 - \lambda(y_j)\}$$

(by definition $S(y_1-) = 1$).

- The *cumulative hazard function* is defined as

$$\Lambda(y_k) = \sum_{1 \leq j \leq k} \lambda(y_j), \quad k \geq 1$$

Proposition

The hazard function characterizes the law of Y . The same is true for the cumulative hazard function.

- **Exercise:** Prove the Proposition.

Hazard functions : continuous random variables (1/2)

- ▶ Assume the random variable $Y \geq 0$ admits the density f
- ▶ The *hazard function* (also called *hazard rate*, or *failure rate*) is defined as

$$\lambda(y) = \frac{f(y)}{S(y)} = \frac{f(y)}{\mathbb{P}(Y \geq y)}, \quad y \geq 0$$

- ▶ Herein we will always use the convention $0/0 = 0!$
- ▶ The *cumulative hazard function* is defined as

$$\Lambda(y) = \int_0^y \lambda(t) dt, \quad y \geq 0.$$

Hazard functions : continuous random variables (2/2)

- In the case of a random variable with a density, which we assume continuous, we also have

$$\lambda(y) = \lim_{h \downarrow 0} \frac{1}{h} \mathbb{P}(Y \in [y, y+h) \mid Y \geq y) \quad (1)$$

Proposition

In the where $Y \geq 0$ admits the density f we have the following relationships

$$S(y) = \exp(-\Lambda(y)) \quad \text{and} \quad f(y) = \lambda(y) \exp\left(-\int_0^y \lambda(t) dt\right).$$

In particular, any of $\lambda(\cdot)$ and $\Lambda(\cdot)$ could be used to characterize the law of Y

Exercises:

- ▶ Prove the relationship (1) in the case where the density $f(\cdot)$ is continuous
- ▶ Prove the Proposition on the previous slide

Mean residual life (1/3)

- ▶ Assume Y is a nonnegative continuous random variable (*i.e.*, it admits a density), with finite mean
- ▶ Let $y \geq 0$ such that $S(y) > 0$
- ▶ The *mean residual life (MRL)* function is defined as the expected value of the remaining duration (or lifetime) after a fixed time point. Formally

$$m(y) = \mathbb{E}(Y - y \mid Y > y) = \frac{\int_y^\infty t f(t) dt}{S(y)} - y$$

- ▶ **Remark:** the following relationship holds true

$$m(y) = \frac{1}{S(y)} \int_y^\infty S(t) dt, \quad y \geq 0 \quad (2)$$

Mean residual life (2/3)

- ▶ Let τ_y a nonnegative random variable with the law given by the relationship

$$\mathbb{P}(\tau_y > t) = \mathbb{P}(Y - y > t \mid Y > y), \quad t \geq 0$$

- ▶ This conditional law model the residual duration at time y given that the initial duration Y is at least equal to y
- ▶ **Exercise:** show that for any y such that $\mathbb{P}(Y > y) > 0$

$$\mathbb{P}(\tau_y > t) = \frac{S(y+t)}{S(y)}$$

and that the map $t \mapsto \mathbb{P}(\tau_y > t)$, $t \geq 0$ is a survival function.

- ▶ **Exercise:** show that we have the relationship $m(y) = \mathbb{E}(\tau_y)$

Mean residual life (3/3): some properties of the MRL

- ▶ $m(0) = \mathbb{E}(Y)$ (this holds true whenever $S(0) = 1$)
- ▶ **Exercise:** Show that

$$m(y) = \int_0^{\infty} \exp\{\Lambda(y) - \Lambda(y+t)\} dt$$

- ▶ When both the hazard function and the MRL exist and MRL is differentiable, we have

$$m'(y) = m(y)\lambda(y) - 1 \quad (3)$$

- ▶ Solving the differential equation, we find the inversion formula

$$S(y) = \frac{m(0)}{m(y)} \exp \left[- \int_0^y \frac{1}{m(t)} dt \right], \quad y \geq 0 \quad (4)$$

- ▶ The function $m(\cdot)$ defined on $\mathbb{R}_+$ characterizes the law of Y

Exercises:

- ▶ Prove the relationships (3) and (4)

Quantiles (1/2)

- ▶ The quantile function of a random variable is a real-valued function defined on $(0, 1)$ defined by the relationship

$$q(p) = \inf\{t : F(t) \geq p\}, \quad 0 < p < 1$$

- ▶ When $Y \geq 0$, the quantile function takes values on $\mathbb{R}_+$
- ▶ We have the properties

$$q(p) = \inf\{t : F(t) > p\} = \inf\{t : S(t) \leq 1 - p\}$$

- ▶ When F is strictly increasing and continuous,

$$q(p) = F^{-1}(p) = S^{-1}(1 - p), \quad 0 < p < 1$$

Quantiles (2/2)

- ▶ $q(0.5)$ is the median, while $q(0.25)$ and $q(0.75)$ are the first and third quartile, respectively.
- ▶ If $q(\cdot)$ is the quantile function of a standard normal law, $q(0.9) \approx 1.28$, $q(0.95) \approx 1.64$ and $q(0.975) \approx 1.96$
- ▶ The quantile function $q(\cdot)$ defined on $(0, 1)$ characterizes the law of a random variable
- ▶ **Remark:** Let Y be a random variable with quantile function $q(\cdot)$. For any $p \in (0, 1)$, $q(p)$ is solution of the optimization problem

$$\min_a \mathbb{E} \{ (Y - a)(p - \mathbf{1}_{\{Y - a < 0\}}) \}.$$

In particular, $q(0.5)$ is solution of the optimization problem

$$\min_a \mathbb{E} \{ |Y - a| \}$$

Exercises:

- Assume that Y is a real valued random variable which admits a density $f(\cdot)$ such that $f(a) > 0, \forall a \in \mathbb{R}$. Let $p \in (0, 1)$ and $q(p) = F^{-1}(p)$.

1. Show that

$$\begin{aligned} \mathbb{E} \{ (Y - q(p))(p - \mathbf{1}_{\{Y - q(p) < 0\}}) \} \\ \leq \mathbb{E} \{ (Y - a)(p - \mathbf{1}_{\{Y - a < 0\}}) \} \quad \forall a \in \mathbb{R} \end{aligned}$$

2. Show that the inequality is strict for any $a \neq q(p)$.

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Duration analysis in presence of covariates

IFR and DFR laws

- ▶ The hazard functions, especially the hazard rate, plays an important role in duration analysis
- ▶ It is common to classify the law of the duration using $\lambda(\cdot)$
- ▶ **Definition (IFR)** A law is said to be an *increasing failure rate (IFR)* distribution if $-\log S(\cdot)$ is convex. When the density exists, this is equivalent to saying that $\lambda(\cdot)$ is nondecreasing
- ▶ **Definition (DFR)** A law is said to be an *decreasing failure rate (DFR)* distribution if $-\log S(\cdot)$ is concave. When the density exists, this is equivalent to saying that $\lambda(\cdot)$ is nonincreasing
- ▶ Other types of shapes (e.g. bathtub-shape) could also be of interest. Other types of classifications could be based on the mean residual life, average hazard functions, *etc*

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Ordering probability laws: stochastic dominance

- ▶ *Stochastic dominance* is a concept which refers to a set of relations that may hold between a pair of probability distributions
- ▶ This concept is quite useful in duration analysis, but also in Economics (for instance, to analyse income distributions and income inequality), in Finance (to compare the returns to various financial assets), *etc*
- ▶ The most common stochastic dominance (or ordering) is the so-called *first-order stochastic dominance*.
- ▶ **Definition:** The probability distribution characterized by the survival function $\bar{F}$ *stochastically dominates* the probability distribution characterized by the survival function $\bar{G}$ *at the order one* if

$$\bar{F}(y) \geq \bar{G}(y), \quad \forall y \quad (\text{or equivalently } F(y) \leq G(y), \quad \forall y)$$

Stochastic dominance: properties

- ▶ Recall notation: if F, G are distribution functions, then $\bar{F}, \bar{G}$ are the corresponding survival functions
- ▶ The concept of stochastic dominance is not restricted to nonnegative variables, it could be defined for real-valued random variables
- ▶ Let $Y \sim F$ and $Y' \sim G$. Under mild conditions, it can be shown that $\bar{F}$ stochastically dominates at the order one $\bar{G}$ if and only if

$$\mathbb{E}[u(Y)] \geq \mathbb{E}[u(Y')]$$

for any nondecreasing function $u(\cdot)$

- ▶ **Exercise:** Let $Y \sim F$ and $Y' \sim G$. Show that if $\bar{F}$ stochastically dominates at the order one $\bar{G}$, then
 1. $\mathbb{E}(Y) \geq \mathbb{E}(Y')$;
 2. if $\mathbb{E}(Y) = \mathbb{E}(Y')$ then necessarily F and G coincide (*here, for simplicity, assume $Y, Y' \geq 0$*);
 3. if Y and Y' are independent, $\mathbb{P}(Y \geq Y') \geq 1/2$.

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Exponential law

- ▶ A law for a nonnegative random variable Y
- ▶ The law is indexed by one positive parameter λ
- ▶ The survivor function: $S(y) = \exp(-\lambda y)$, $y \geq 0$
- ▶ Density: $f(y) = \lambda \exp(-\lambda y)$, $y \geq 0$
- ▶ $\mathbb{E}(Y) = 1/\lambda$; $\text{Var}(Y) = 1/\lambda^2$
 - ▶ Some authors use a different parametrization: $\gamma = 1/\lambda$!
- ▶ Mode: one mode at $y = 0$
- ▶ Quantile function: $q(p) = -\lambda^{-1} \log(1 - p)$
- ▶ Hazard function: $\lambda(y) \equiv \lambda$ (constant hazard rate)
- ▶ Cumulative Hazard function: $\Lambda(y) = \lambda y$
- ▶ Mean residual life: $m(y) = 1/\lambda$ (lack of memory)
- ▶ Simultaneously IFR and DFR

Weibull law

- ▶ A law for a nonnegative random variable Y
- ▶ The law is indexed by two parameters $\lambda > 0$ (scale parameter) and $k > 0$ (shape parameter)
- ▶ The survivor function: $S(y) = \exp(-(\lambda y)^k)$, $y \geq 0$
- ▶ Density: $f(y) = k\lambda(\lambda y)^{k-1} \exp(-(\lambda y)^k)$, $y \geq 0$
- ▶ $\mathbb{E}(Y) = \Gamma(1 + 1/k)/\lambda$; $\text{Var}(Y)$ – exercise
 - ▶ Some authors use a different parametrization:
 $\gamma = 1/\lambda$, or $\gamma = \lambda^k$, or yet other !
- ▶ The random variable $W = (\lambda Y)^k$ has an exponential law with parameter equal to 1
- ▶ If $U \sim \mathcal{U}[0, 1]$, then $Y = \lambda^{-1}(-\log(U))^{1/k}$ is a Weibull random variable with parameters λ and k
- ▶ One mode at $y = 0$ if $k \leq 1$, and at $((k-1)/k)^{1/k}/\lambda$ if $k > 1$
- ▶ Quantile function: $q(p) = \lambda^{-1}(-\log(1-p))^{1/k}$
- ▶ Hazard function: $\lambda(y) = k\lambda(\lambda y)^{k-1}$
- ▶ IFR if $k \geq 1$ and DFR if $k \leq 1$

Modeling the duration law

- ▶ Use **parametric** families (exponential, Weibull, *etc*)
- ▶ In the presence of the covariates there are many possible strategies. For instance let the parameters of a Weibull law depending on the covariates and some regression parameters.
- ▶ It is also possible to estimate the survival function **nonparametrically**. Specific nonparametric methods for the hazard rate were also proposed.
- ▶ Such nonparametric procedure could be adapted to include covariates, but they suffer from the *curse of dimensionality*
- ▶ Common approaches propose to combine the flexibility of the nonparametric methods with the appealing features of parametric modeling
→ **semiparametric** approaches in the presence of covariates (proportional hazard, proportional odds, . . .)
- ▶ These aspects will be elaborated in the following

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Types of censoring

Truncation

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Duration are often censored

- ▶ A specificity of the duration analysis is that durations are not always observed precisely
- ▶ This aspect will require a specific treatment in the statistical inference
- ▶ *Censoring* is perhaps the most common reason of incomplete observation of the duration
- ▶ We could distinguish
 - ▶ deterministic/random censoring
 - ▶ left/right censoring
 - ▶ but also (not studied herein...)
 - ▶ left and right censoring
 - ▶ interval censoring

Type I and type II censoring (1/2)

- ▶ **Type I censoring:** the event is observed only if it occurs prior to some pre-specified time (that may vary from individual to individual)
 - ▶ data consists of the pairs of random variables (T, δ) where

$$T_i = Y_i \wedge C_i \quad \text{and} \quad \delta_i = \mathbf{1}_{\{Y_i \leq C_i\}}, \quad 1 \leq i \leq n$$

- ▶ **A generalization of Type I censoring:** when individuals enter the study at different times and the terminal point of the study is predetermined by the investigator, so that the censoring times are known when an individual is entered into the study.

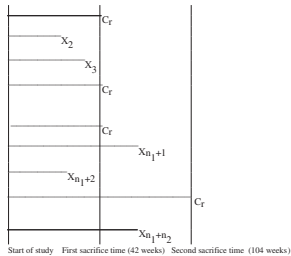


Figure 3.2 *Type I censoring with two different sacrifice times*

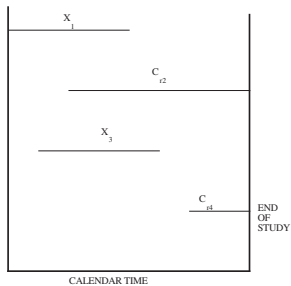


Figure 3.3 *Generalized Type I censoring when each individual has a different starting time*

Type I and type II censoring (2/2)

- ▶ **Type II censoring:** the study continues until the failure of the first r individuals, where r is some predetermined integer, smaller than n
- ▶ Type II censoring occurs often in testing of equipment life
- ▶ **Progressive Type II censoring:** the first r_1 failures in a sample of n units are noted and recorded. Then $n_1 - r_1$ of the remaining $n - r_1$ unfailed units are removed from the experiment (leaving $n - n_1$ units on study). When the next r_2 units (another integer chosen prior to the start of the study) fail, $n_2 - r_2$ of the unfailed units are removed. This process continues until some predecided series of repetitions is completed. The integers $r_1, n_1, r_2, n_2, \dots$, are fixed integers, chosen prior to the start of the study!

Random censoring (1/2)

- ▶ Type I and Type II censoring and their extensions are right-censoring schemes: the durations not observed are larger than some value known or observed by the statistician
- ▶ A very common right-censoring scheme is the so-called *random right-censoring*. When no possible confusion, we simply call it *right-censoring*
 - ▶ There exists a **random variable** C (the right censoring) and data consists of the pairs of random variables (T, δ) where

$$T_i = Y_i \wedge C_i \quad \text{and} \quad \delta_i = \mathbf{1}_{\{Y_i \leq C_i\}}, \quad 1 \leq i \leq n$$

- ▶ Even if the observations seem to be obtained in a similar way in various censoring schemes, for the inference purpose it is very important to distinguish deterministic and random censoring !!

Random censoring (2/2)

- ▶ **Left-censoring:** a duration is left censored if it is only known to be less than some observed value, usually considered to be randomly generated
 - ▶ There exists a **random variable** L (the left censoring) and data consists of the pairs of random variables (T, δ) where

$$T_i = Y_i \vee L_i \quad \text{and} \quad \delta_i = \mathbf{1}_{\{Y_i \geq L_i\}}, \quad 1 \leq i \leq n$$

- ▶ **Left and right-censoring:** a duration Y is left and right-censored if it is precisely observed only when $L \leq Y \leq R$, where L and R are random left and right censoring times, such that $\mathbb{P}(L \leq R) = 1$. The data consists of the pairs of random variables (T, Δ) where

$$T_i = \max(\min(Y_i, R_i), L_i) = \min(\max(Y_i, L_i), R_i), \quad 1 \leq i \leq n$$

and Δ_i indicated which of situations $T_i = Y_i$, $T_i = L_i$ or $T_i = R_i$ is observed.

Ignoring the censoring induces biases (1/2)

- ▶ Consider the case of a random right-censoring
- ▶ The variable of interest is $Y \geq 0$ and the observed duration is $T = Y \wedge C$, with $C \geq 0$; imagine that we have i.i.d. observations
- ▶ Imagine that we want to estimate $\theta = \mathbb{E}(Y)$ under the assumption $Y \perp C$ (untestable assumption!)
- ▶ **Bad idea 1:** take

$$\hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n T_i$$

Then, since by the independence assumption $S_T = S_Y S_C$,

$$\mathbb{E}[\hat{\theta}_1] = \mathbb{E}[T] = \int_0^\infty S_Y(t) S_C(t) dt \leq \int_0^\infty S_Y(t) dt = \mathbb{E}[Y]$$

and in general the inequality is strict

Ignoring the censoring induces biases (2/2)

- ▶ Consider the same problem, under the same assumptions
- ▶ **Bad idea 2:** given the sample of $\delta = \mathbf{1}_{\{Y \leq C\}}$, take

$$\hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n \delta_i T_i$$

- ▶ Then, since by the independence assumption
 $\mathbb{E}(\delta \mid Y) = S_C(Y-)$, and noting that $\delta T = \delta Y$,

$$\mathbb{E}[\hat{\theta}_2] = \mathbb{E}[\delta T] = \mathbb{E}[\delta Y] = \mathbb{E}[\mathbb{E}(\delta \mid Y) Y] = \mathbb{E}[S_C(Y-) Y] \leq \mathbb{E}[Y],$$

and in general the inequality is strict

- ▶ **Good idea:** find a weight $\omega(T, \delta)$, function of the observations (T, δ) such that

$$\mathbb{E}[\omega(T, \delta) T] = \mathbb{E}[Y].$$

- ▶ The choices $\omega(\cdot) \equiv 1$ and $\omega(T, \delta) = \delta$ do not work...
- ▶ Any suggestion?
- ▶ Is your suggestion feasible? Elaborate a bit.
- ▶ Would it automatically work for other type of censoring?

- ▶ Could you connect this with the propensity score in missing data situations, under the MAR assumption?

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Truncation is not censoring

- ▶ A second type of incompleteness in duration data is induced by the *truncation*
- ▶ Truncation is different from censoring! There is more information with censoring.
- ▶ Truncation of duration occurs when only those individuals whose time-to-event lies within a certain observational window $[\tau_L, \tau_R]$ (deterministic or random) are observed.
- ▶ An individual whose event time is not in this interval is not observed and no information on this subject is available
- ▶ When $\tau_R = \infty$ we have left-truncation, while $\tau_L = 0$ yields the right-truncation
- ▶ Because the statistician is only aware of individuals with event times in the observational window, the inference for truncated data is restricted to the conditional law of the duration given that the duration is in the observation window

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Recall the likelihood principle

- ▶ Consider a model, *i.e.* a family of probability distributions
- ▶ Assume that the laws admit a density (w.r.t. some dominating measure); typically
 - ▶ the counting measure for models for discrete variables
 - ▶ the Lebesgue measure for models for continuous variables
- ▶ Let $\mathcal{M}$ denote the family (the set) of densities in the model
- ▶ Given the i.i.d. sample $Y_1, \dots, Y_n$, the likelihood function is defined as

$$\mathcal{L}_n(f) = \prod_{i=1}^n f(Y_i), \quad f \in \mathcal{M}$$

- ▶ In a parametric approach, $\mathcal{M} = \{f_\theta : \theta \in \Theta\}$ and the likelihood becomes

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n f_\theta(Y_i), \quad \theta \in \Theta$$

- ▶ The maximum likelihood estimator is a maximizer of $\mathcal{L}_n(\theta)$ (or $\mathcal{L}_n(f)$)

The likelihood under censoring/truncation

- The contributions to the likelihood

observation	likelihood contribution
exact (see above)	$f(y)$
right-censored	$S(C)$
left-censored	$F(L)$
interval-censored	$S(L) - S(R)$
left-truncated	$f(y)/S(\tau_L)$
right-truncated	$f(y)/\{1 - S(\tau_R)\}$
interval-truncated	$f(y)/\{S(\tau_L) - S(\tau_R)\}$

- The likelihood function may be constructed by putting together the component parts. For instance, *with censoring*

$$\mathcal{L}_n(f) \propto \prod_{i \in D} f(Y_i) \prod_{i \in L} F(L_i) \prod_{i \in R} S(C_i) \prod_{i \in I} \{S(L_i) - S(R_i)\}$$

where

- D is the set of death times, R the set of right-censored observations, L the set of left-censored observations, and I the set of interval-censored observations.

The likelihood under random right-censoring (1/2)

- ▶ The duration of interest is Y
- ▶ The data consist of independent realizations $(T_1, \delta_1), \dots, (T_n, \delta_n)$ of (T, δ) where
 - ▶ $\delta \in \{0, 1\}$
 - ▶ $T = Y$ if $\delta = 1$ and $T < Y$ if $\delta = 0$
- ▶ Usually, the Statistician considers that there exists a random duration C , the right-censoring, such that

$$T = \min(Y, C) = Y \wedge C \quad \text{and} \quad \delta = \mathbf{1}_{\{Y \leq C\}}$$

- ▶ Two convenient, *but critical* assumption
 - ▶ Y and C are independent (*independent censoring*)
 - ▶ C does not carry any information on the law of Y (*non informative censoring*)

The likelihood under random right-censoring (2/2)

► Let

- f and $\bar{F}$ be the density and the survival function of Y
- g and $\bar{G}$ be the density and the survival function of C

(Remark: the dominating measure needs not to be the Lebesgue measure)

Proposition

Under the assumption of independent censoring, the contribution of observation (T_i, δ_i) to the likelihood is

$$\left\{ f(T_i) \bar{G}(T_i-) \right\}^{\delta_i} \left\{ \bar{F}(T_i) g(T_i) \right\}^{1-\delta_i} \quad \left(\text{or} \quad \left\{ f_{\theta}(T_i) \bar{G}(T_i-) \right\}^{\delta_i} \left\{ \bar{F}_{\theta}(T_i) g(T_i) \right\}^{1-\delta_i} \right)$$

If in addition the censoring is non informative, the contribution to the likelihood that matters for inference on the law of Y is

$$f(T_i)^{\delta_i} \bar{F}(T_i)^{1-\delta_i} \quad \left(\text{or} \quad f_{\theta}(T_i)^{\delta_i} \bar{F}_{\theta}(T_i)^{1-\delta_i} \right)$$

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Parametric likelihood inference under random right-censoring (1/3)

- ▶ Consider an exponential model for Y
- ▶ Recall (here $\theta = \lambda$)

$$S_\lambda(y) = \exp(-\lambda y), \quad f_\lambda(y) = \lambda \exp(-\lambda y), \quad y \geq 0$$

- ▶ Then, with the data (T_i, δ_i) , $1 \leq i \leq n$, the likelihood is

$$\mathcal{L}_n(\lambda) = \prod_{i=1}^n f_\lambda(T_i)^{\delta_i} \bar{F}_\lambda(T_i)^{1-\delta_i} = \lambda^{\sum_{i=1}^n \delta_i} \exp\left(-\lambda \sum_{i=1}^n T_i\right), \quad \lambda > 0$$

- ▶ The log-likelihood is

$$\begin{aligned} \log \mathcal{L}_n(\lambda) &= \sum_{i=1}^n \{\delta_i \log f_\lambda(T_i) + (1 - \delta_i) \log \bar{F}_\lambda(T_i)\} \\ &= \log(\lambda) \sum_{i=1}^n \delta_i - \lambda \sum_{i=1}^n T_i, \quad \lambda > 0 \end{aligned}$$

Exercises:

- ▶ **Exercise:** derive the maximum likelihood estimator (MLE) in the exponential model. Show that it is consistent. What condition is needed on g or $\overline{G}$?
 - ▶ *Hint:* Compute separately $\mathbb{E}(\delta) = \mathbb{P}(Y \leq C)$ and $\mathbb{E}(T) = \mathbb{E}(Y \wedge C)$. Next, use the law of large numbers to deduce that the MLE converges to the ratio of these two quantities. The ratio is equal to the true value of λ .
- ▶ **Exercise:** Show that the (MLE) in the exponential model is $\sqrt{n}$ -asymptotically normal.

- ▶ **Remark:** In the usual parametric models used in duration models, it could be shown that the MLE is consistent and $\sqrt{n}$ -asymptotically normal. However, they asymptotic law depend on the law of the censoring variable C !!
- ▶ In general, tests and confidence intervals for the parameter θ could not be realized without additional assumptions or procedures

Parametric likelihood inference under random right-censoring (3/3)

- ▶ Consider a Weibull model for Y
- ▶ Recall (here $\theta = (\lambda, k)'$)

$$S_{\lambda,k}(y) = \exp(-(\lambda y)^k), \quad f_{\lambda,k}(y) = k\lambda(\lambda y)^{k-1} \exp(-(\lambda y)^k),$$

- ▶ Then, with the data (T_i, δ_i) , $1 \leq i \leq n$, the log-likelihood is

$$\log \mathcal{L}_n(\lambda, k) = \sum_{i=1}^n \delta_i \left\{ \log(\lambda^k k) + (k-1) \log(T_i) \right\} - \lambda^k \sum_{i=1}^n T_i^k, \quad \lambda, k > 0$$

- ▶ The MLE does not have an explicit form, but it is still convergent and $\sqrt{n}$ -asymptotically normal.

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The likelihood under left-truncation (1/2)

- ▶ Consider the situation of left-truncated duration
- ▶ The duration of interest and the truncation time are assumed independent
- ▶ The data are the i.i.d. pairs $(Y_i, \tau_{L,i})$, $1 \leq i \leq N$, where here N is **random**
- ▶ There are n individuals, with n *unknown*, and only the individuals with $Y_i \geq \tau_{L,i}$ are observed! Hence, the number N of individuals with this property is random.
- ▶ In such cases, one could follow the conditional likelihood approach

The likelihood under left-truncation (2/2)

- ▶ With a parametric approach, the conditional likelihood given N and the observed values $\tau_{L,i}$ is

$$\mathcal{L}_n(\theta) = \prod_{i=1}^N \frac{f_{\theta}(Y_i)}{S(\tau_{L,i})}, \quad \theta \in \Theta$$

- ▶ If the law of the truncation variable is given (or modeled), the parametric conditional likelihood given N is

$$\begin{aligned} \mathcal{L}_n(\theta) &= \left(\int \mathbf{1}_{y \geq t} f(y) \xi(t) dy dt \right)^{-N} \prod_{i=1}^N f_{\theta}(Y_i) \xi(\tau_{L,i}) \\ &= \left(\int f(y) \Psi(y) dy \right)^{-N} \prod_{i=1}^N f_{\theta}(Y_i) \xi(\tau_{L,i}), \quad \theta \in \Theta \end{aligned}$$

where $\xi(\cdot)$ is the density of τ_L and $\Psi(\cdot)$ is the corresponding distribution function

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Nonparametric model

- ▶ To avoid misspecification, one would like to consider a large model $\mathcal{M}$
- ▶ Why not all the possible laws for Y in the model? Then the model would be a *nonparametric* one.
- ▶ Nonparametric models are appealing because they avoid misspecification, but sometimes are too complex and difficult to handle
- ▶ What would become the MLE in such a case? It is easy to see that the problem is degenerate if one wants to maximize over sets of all densities
- ▶ To address these questions **we will focus on the case of duration possibly right-censored**
- ▶ The nonparametric MLE exists, in some sense, but there are simpler ways to introduce a nonparametric estimate of the law of the duration of interest under random right-censoring

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Empirical distribution function

- ▶ Let $Y_1, \dots, Y_n$ a sample with a same law
- ▶ The *empirical distribution function* is defined as

$$\hat{F}(y) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{Y_i \leq y\}}, \quad y \in \mathbb{R}.$$

It is a sufficient statistics. It is constant between the observed values.

- ▶ It is an unbiased estimator of the distribution function $F(\cdot)$
- ▶ If the observations are independent, it is quite easy
 - ▶ to calculate the variance
 - ▶ to prove the consistency
 - ▶ to prove the $\sqrt{n}$ -asymptotic normality
- ▶ These properties holds for discrete, continuous or more general random variables Y

- ▶ It is also easy to check that applied with $T_i = Y_i \wedge C_i$, the empirical distribution function is no longer an unbiased and consistent estimator of $F(\cdot)$. Exercise!
 - ▶ *Hint:* try to show the statement for the survival function instead, next deduce it for the distribution function
- ▶ One has to correct the bias due to the censored observations, that is due to T_i that are equal to C_i and strictly smaller than Y_i
- ▶ We do this in the following
 - ▶ First, we try to understand some aspects in the case without censoring
 - ▶ Next we go back to the situation of interest in this course, the case with (right-)censoring

Empirical distribution and the hazard function (1/2)

- ▶ Recall that for a discrete law with ordered support $\{y_1, y_2, \dots\}$ we have

$$\lambda(y_k) = \frac{p_k}{S(y_{k-})} \quad (\text{by definition } S(y_1-) = 1)$$

and

$$S(y_k) = \prod_{j=1}^k \{1 - \lambda(y_j)\}$$

- ▶ The probabilities p_k and $S(y_{k-})$ could be easily estimated by the observed frequencies
- ▶ We will use this idea to rewrite the empirical distribution function

Empirical distribution and the hazard function (2/2)

- ▶ Let $y_1, y_2, \dots$ be the **ordered distinct** values in the sample $Y_1, \dots, Y_n$. Ties (*ex-echos*) are allowed

- ▶ Define

$$\hat{\lambda}(y_k) = \frac{\sum_{j=1}^n \mathbf{1}_{\{Y_j=y_k\}}}{\sum_{j=1}^n \mathbf{1}_{\{Y_j \geq y_k\}}}$$

- ▶ Then, for any y ,

$$\hat{S}(y) = 1 - \hat{F}(y) = \prod_{\{k: y_k \leq y\}} \{1 - \hat{\lambda}(y_k)\} = \prod_{\{k: y_k \leq y\}} \left\{ 1 - \frac{\sum_{j=1}^n \mathbf{1}_{\{Y_j=y_k\}}}{\sum_{j=1}^n \mathbf{1}_{\{Y_j \geq y_k\}}} \right\}$$

(By convention, the product with an empty set of indices is equal to 1.)

- ▶ In the case with *no ties* in the sample of size n , one can write

$$\hat{S}(y) = 1 - \hat{F}(y) = \prod_{\{i: Y_{(i)} \leq y\}} \left\{ 1 - \frac{1}{n - i + 1} \right\} = \prod_{\{i: Y_{(i)} \leq y\}} \frac{n - i}{n - i + 1}$$

where $Y_{(1)}, Y_{(2)}, \dots, Y_{(n)}$ is the order statistics, that is the values of the sample (increasingly) ordered

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Kaplan-Meier estimator (1/3)

- ▶ The sample is (T_i, δ_i) , $1 \leq i \leq n$, where $T_i = Y_i \wedge C_i$ and $\delta_i = \mathbf{1}_{\{Y_i \leq C_i\}}$
- ▶ Let $t_1, t_2, \dots$ be the **ordered distinct** values in the sample $T_1, \dots, T_n$. **Ties (ex-echos) are allowed**
- ▶ The key point is the definition of the hazard function. Under right-censoring, the suitable definition is

$$\hat{\lambda}(t_k) = \frac{\sum_{j=1}^n \delta_j \mathbf{1}_{\{T_j = t_k\}}}{\sum_{j=1}^n \mathbf{1}_{\{T_j \geq t_k\}}} = \frac{\#\{\text{failures at time } t_k\}}{\#\{\text{individuals 'at risk' at time } t_k\}}, \quad k = 1, 2, \dots$$

- ▶ Then, $\forall y$, the estimator of the survivor at y is

$$\hat{S}(y) = 1 - \hat{F}(y) = \prod_{\{k: t_k \leq y\}} \{1 - \hat{\lambda}(t_k)\} = \prod_{\{k: t_k \leq y\}} \left\{ 1 - \frac{\sum_{j=1}^n \delta_j \mathbf{1}_{\{T_j = t_k\}}}{\sum_{j=1}^n \mathbf{1}_{\{T_j \geq t_k\}}} \right\}$$

The function $y \mapsto \hat{S}(y)$ is the Kaplan-Meier estimator of $S(\cdot)$

Arguments for the hazard function estimator

- ▶ Let us try to understand the definition of $\hat{\lambda}(t_k)$ on the previous slide in the case of a discrete durations
- ▶ Using the law of large numbers, for any t_k , we can write

$$\begin{aligned}\hat{\lambda}(t_k) &= \frac{n^{-1} \sum_{j=1}^n \delta_j \mathbf{1}_{\{T_j=t_k\}}}{n^{-1} \sum_{j=1}^n \mathbf{1}_{\{T_j \geq t_k\}}} \approx \frac{\mathbb{E}(\delta \mathbf{1}_{\{T=t_k\}})}{\mathbb{E}(\mathbf{1}_{\{T \geq t_k\}})} \\ &= \frac{\mathbb{E}(\delta \mathbf{1}_{\{Y=t_k\}})}{S_Y(t_k-) S_C(t_k-)} = \frac{\mathbb{E}[\mathbb{E}(\delta \mid Y) \mathbf{1}_{\{Y=t_k\}}]}{S_Y(t_k-) S_C(t_k-)} \\ &= \frac{\mathbb{E}[S_C(Y-) \mathbf{1}_{\{Y=t_k\}}]}{S_Y(t_k-) S_C(t_k-)} = \frac{\mathbb{P}(Y = t_k)}{\mathbb{P}(Y \geq t_k)} = \lambda(t_k)\end{aligned}$$

- ▶ We used the independence between C and Y and the following identity: for any real-valued function $\phi(\cdot)$,

$$\delta \phi(T) = \delta \phi(Y)$$

Kaplan-Meier estimator (2/3)

- ▶ The Kaplan-Meier estimator has jumps (puts positive mass) only at the non-censored observations
- ▶ When there are no ties, the Kaplan-Meier estimator could be rewritten, $\forall y$

$$\hat{S}(y) = 1 - \hat{F}(y) = \prod_{\{k: T_k \leq y\}} \left\{ 1 - \frac{1}{\sum_{j=1}^n \mathbf{1}_{\{T_j \geq T_k\}}} \right\}^{\delta_k} \quad (5)$$

- ▶ If, in addition, the sample (T_i, δ_i) , $1 \leq i \leq n$, is given such that the T_i are ordered increasingly, then the Kaplan-Meier estimator could be rewritten, $\forall y$

$$\hat{S}(y) = 1 - \hat{F}(y) = \prod_{\{k: T_k \leq y\}} \left\{ 1 - \frac{1}{n - k + 1} \right\}^{\delta_k}$$

Kaplan-Meier estimator (3/3)

- ▶ Kaplan-Meier (KM) estimator is an appropriate estimator of $S(\cdot)$ under random right-censoring
- ▶ The jumps of the Kaplan-Meier estimator are not multiples of $1/n$. They have some interpretation (see below)
- ▶ When the last observation is uncensored, the KM estimator is a survivor function (it vanishes from the last observation to the right).
- ▶ When the last observation is censored, the KM estimator does not vanish, it is not a proper survival function. The common convention is to say that it has a mass at infinity.
- ▶ One could use the KM estimator for estimating the survival function of the censoring times C_i
 - ▶ In the case with no ties, one has only to replace δ_k by $1 - \delta_k$ in formula (5) above
 - ▶ The case with ties is slightly more elaborate
- ▶ The Kaplan-Meier estimator is a kind of *plug-in* estimator

Nelson-Aalen estimator (1/3)

- ▶ Let $t_1, t_2, \dots$ be the **ordered distinct** values in the sample $T_1, \dots, T_n$. **Ties (*ex-echos*) are allowed**
- ▶ We defined the empirical counterpart of the hazard function, *in the presence of right-censoring*, as

$$\hat{\lambda}(t_k) = \frac{\sum_{j=1}^n \delta_j \mathbf{1}_{\{T_j=t_k\}}}{\sum_{j=1}^n \mathbf{1}_{\{T_j \geq t_k\}}} = \frac{\#\{\text{failures at time } t_k\}}{\#\{\text{individuals 'at risk' at time } t_k\}}, \quad k = 1, 2, \dots$$

- ▶ Then the natural estimator of $\Lambda(y)$, the value of the cumulative hazard function at point y , is

$$\hat{\Lambda}(y) = \sum_{\{k: t_k \leq y\}} \hat{\lambda}(t_k).$$

- ▶ The function $\hat{\Lambda}(\cdot)$ is the so-called *Nelson-Aalen* estimator of $\Lambda(\cdot)$

Nelson-Aalen estimator (2/3)

- ▶ We could also rewrite it as

$$\hat{\Lambda}(y) = \sum_{\{k: T_k \leq y\}} \hat{\lambda}(T_k)$$

- ▶ If the sample (T_i, δ_i) , $1 \leq i \leq n$, has no ties and the T_i are ordered increasingly, then the Nelson-Aalen (N-A) estimator could be rewritten, $\forall y$

$$\hat{\Lambda}(y) = \sum_{\{k: T_k \leq y\}} \frac{\delta_k}{n - k + 1}$$

- ▶ Like the KM estimator, the N-A estimator has jumps *only* at the uncensored observations (the ones with $\delta_k = 1$)

Nelson-Aalen estimator (3/3) – properties

- ▶ The jump of the N-A at the observation T_k (when no ties in the durations and the T_i 's are ordered increasingly) is

$$\hat{\Lambda}(\{T_k\}) = \hat{\Lambda}(T_k) - \hat{\Lambda}(T_k-) = \frac{\delta_k}{n - k + 1} \in [0, 1]$$

- ▶ Meanwhile (no ties and ↗), the jump of the KM estimator at **an uncensored** T_k is

$$\hat{S}(\{T_k\}) = \hat{S}(T_k-) - \hat{S}(T_k) = \frac{1}{n - k + 1} \prod_{1 \leq j < k} \left\{ 1 - \frac{1}{n - j + 1} \right\}^{\delta_j}$$

- ▶ If the last observation is uncensored, the jump of the N-A estimator is 1 and the KM estimator is a proper survival function (it vanishes after the last observation)
- ▶ Otherwise, the jump of the N-A estimator is strictly smaller than 1 and the KM is not a proper survival function (it has a plateau above zero).
- ▶ The N-A estimator is also a kind of *plug-in* estimator

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Asymptotics for KM and N-A estimators (1/8)

- ▶ For many statistical models, there are two major types of asymptotic results
 - ▶ Consistency – usually derived from the Law of Large Numbers (LLN)
 - ▶ $\sqrt{n}$ –asymptotic normality – usually derived from the Central Limit Theorem (CLT)
- ▶ **Law of Large Numbers – no censoring case:** Let $\hat{F}(\cdot)$ be the empirical distribution function of an i.i.d. sample from a random variable Y with distribution function $F(\cdot)$ on the real line. Then, for any function $\phi(\cdot)$ with $\mathbb{E}(|\phi(Y)|) < \infty$,

$$\frac{1}{n} \sum_{i=1}^n \phi(Y_i) = \int \phi(y) \hat{F}(dy) \rightarrow \int \phi(y) F(dy) = \mathbb{E}(\phi(Y)), \quad \text{in probability}$$

Convergence is even almost surely... let's do it simple!

- ▶ The previous convergence holds even *uniformly* over classes of functions satisfying some mild conditions (see the Glivenko-Cantelli classes of functions) – beyond our scope!

- ▶ Some usual functions $\phi(\cdot)$
 - ▶ $\phi(y) = \mathbf{1}_{\{y \leq t\}}$ (resp. $= \mathbf{1}_{\{y > t\}}$) $\rightarrow$ estimate $F(t)$ (resp. $S(t)$)
 - ▶ $\phi(y) = y^p \rightarrow$ estimate the p th moment of Y
 - ▶ $\phi(y) = (y - t)\mathbf{1}_{\{y > t\}}(1 - F(t))^{-1}$ for some t fixed such that $F(t) < 1 \rightarrow$ estimate the MRL function of Y at t
- ▶ We know the **Glivenko-Cantelli Th.** (no censoring case):

$$\sup_{y \in \mathbb{R}} |\hat{S}(y) - S(y)| \rightarrow 0 \quad \left(\text{and} \quad \sup_{y \in \mathbb{R}} |\hat{F}(y) - F(y)| \rightarrow 0 \right), \quad \text{in probability}$$

- ▶ Note that Glivenko-Cantelli Theorem implies the uniform convergence of the empirical cumulative hazard function defined previously in the lecture
- ▶ Do we have similar results for Kaplan-Meier and Nelson-Aalen estimators when the durations are right-censored?

Asymptotics for KM and N-A estimators (3/8)

Theorem (Glivenko-Cantelli Theorem for the KM and N-A)

Consider $(Y_1, C_1), \dots, (Y_n, C_n)$ i.i.d. realizations of Y and C with distribution functions $F(\cdot)$ and $G(\cdot)$, respectively. Assume that Y and C are independent. The observed sample is $T_i = Y_i \wedge C_i$ and $\delta_i = \mathbf{1}_{\{Y_i \leq C_i\}}$, $1 \leq i \leq n$. Then,

$$\sup_{y \in \mathcal{T}} \left| \hat{S}(y) - S(y) \right| \rightarrow 0 \quad \text{and} \quad \sup_{y \in \mathcal{T}} \left| \hat{\Lambda}(y) - \Lambda(y) \right| \rightarrow 0 \quad \text{in probability,}$$

where

- ▶ here $\hat{S}(\cdot)$ is the Kaplan-Meier estimator of $S(\cdot)$ and $\hat{\Lambda}(\cdot)$ is the Nelson-Aalen estimator of $\Lambda(\cdot)$, the cumulative hazard function of Y ;
- ▶ $\mathcal{T} = \{y : F(y-) < 1, G(y-) < 1\}$.

W. Stute & J.L. Wang (1993, *Annals of Statistics*) – showed the a.s. convergence version of this result

- ▶ **Central Limit Theorem – no censoring case:** Let $\hat{F}(\cdot)$ be the empirical distribution function of an i.i.d. sample from a random variable Y with distribution function $F(\cdot)$ on the real line. Then, for any function $\phi(\cdot)$ such that $\mathbb{E}[\phi^2(Y)] < \infty$,

$$\sqrt{n} \left(\int \phi(y) \hat{F}(dy) - \int \phi(y) F(dy) \right) \rightsquigarrow N(0, \text{Var}(\phi(Y)))$$

- ▶ With $\phi(y) = \mathbf{1}\{y > t\}$,

$$\sqrt{n} \left(\hat{S}(t) - S(t) \right) \rightsquigarrow N(0, F(t)S(t))$$

- ▶ Pointwise *asymptotic confidence intervals* could be built !
 - ▶ *Confidence bands* is a different matter !! A stronger version of the CLT allows to built also *confidence bands*
→ beyond our scope!
- ▶ Do we have similar results for Kaplan-Meier and Nelson-Aalen estimators when the durations are right-censored?

Theorem (CLT for the KM and N-A)

Consider

- ▶ $(Y_1, C_1), \dots, (Y_n, C_n)$ i.i.d. realizations of Y and C , independent with distribution functions $F(\cdot)$ and $G(\cdot)$, respectively;
- ▶ the data are $T_i = Y_i \wedge C_i$ and $\delta_i = \mathbf{1}_{\{Y_i \leq C_i\}}$, $1 \leq i \leq n$.

Let y such that $F(y) < 1$ and $G(y-) < 1$. Then,



$$\sqrt{n} \left(\hat{\Lambda}(y) - \Lambda(y) \right) \rightsquigarrow N(0, V^2(y)) \quad \text{with} \quad V^2(y) = \int_0^y \frac{F(dt)}{S^2(t-) \overline{G}(t-)}$$



$$\sqrt{n} \left(\hat{S}(y) - S(y) \right) \rightsquigarrow N(0, S^2(y) V^2(y)),$$

where here

- ▶ $\hat{S}(\cdot)$ is the KM estimator of $S(\cdot)$;
- ▶ $\hat{\Lambda}(\cdot)$ is the N-A estimator of the cumulative hazard function.

Asymptotics for KM and N-A estimators (6/8)

- ▶ If not stated differently, we will consider Y and C continuous random variables, in which case we could rewrite $V^2(y)$ under the form

$$V^2(y) = \int_0^y \frac{F(dt)}{S^2(t)\overline{G}(t)} \quad \text{or} \quad V^2(y) = \int_0^y \frac{f(t)dt}{S^2(t)\overline{G}(t)}$$

where $f(\cdot)$ is the density of Y

- ▶ To be really useful, the asymptotic variance should be estimated. It could be done using
 - ▶ resampling → this will be the topic in another lecture... KM could be an example of application
 - ▶ estimates of the quantities appearing in the variance expression
- ▶ For estimating the variance, one could observe that $S(t)\overline{G}(t) = \mathbb{P}(T > t)$, and this probability could be easily estimated from data!

- ▶ The approximate variance of the KM estimator is

$$\frac{1}{n} S^2(y) V^2(y)$$

- ▶ One could estimate $S^2(y)$ by $\hat{S}^2(y)$
- ▶ Note that we could write

$$V^2(y) = \int_0^y \frac{1}{\mathbb{P}(\bar{T} > t)} \Lambda(dt)$$

and thus estimate $V^2(y)$ by

$$\hat{V}^2(y) = n \sum_{\{k: T_k \leq y\}} \frac{\hat{\Lambda}(\{T_k\})}{\sum_{j=1}^n \mathbf{1}_{\{T_j > T_k\}}},$$

where $\hat{\Lambda}(\{T_k\})$ is the jump of the N-A estimator at T_k .

Asymptotics for KM and N-A estimators (8/8)

- ▶ Asymptotic confidence interval of $S(y)$ (of nominal level α) is then

$$IC_a(\alpha) = \left[\hat{S}(y) \pm \frac{z_{\alpha/2} \hat{S}(y)}{\sqrt{n}} \sqrt{\hat{V}^2(y)} \right]$$

where $z_{\alpha/2}$ is the $(\alpha/2)$ -th quantile of the $N(0, 1)$ law

- ▶ With finite samples, the confidence interval $IC_a(\alpha)$ behaves poorly when $\hat{S}(y)$ is close to 0 or 1; better versions have been proposed and implemented
- ▶ **Exercise:** propose an estimator for the asymptotic variance of the N-A estimator at any y such that $F(y) < 1$ and $G(y-) < 1$.
- ▶ **Exercise*:** what becomes the estimator $n^{-1} \hat{S}^2(y) \hat{V}^2(y)$ when there is no censoring? Compare it with the simple estimator $n^{-1} \hat{S}(y) \hat{F}(y)$ one would use when data are not censored.

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The jumps of the KM and IPCW (1/3)

- ▶ Under random right-censoring, to estimate $\mathbb{E}(\phi(Y))$ for some function $\phi(\cdot)$ with $\mathbb{E}(|\phi(Y)|) < \infty$, one could use the KM estimator
- ▶ More precisely, one could estimate $\int \phi(y)F(dy) = \mathbb{E}(\phi(Y))$ by the *Kaplan-Meier integral*

$$\sum_{i=1}^n \phi(T_i) \hat{S}(\{T_i\}) = - \int \phi(y) \hat{S}(dy)$$

where

- ▶ $\hat{S}(\cdot)$ is the Kaplan-Meier estimator of $S(\cdot)$;
- ▶ $\hat{S}(\{T_i\})$ is the jump (the positive mass) at the T_i , that means $\hat{S}(\{T_i\}) \stackrel{\text{def}}{=} \hat{S}(T_i-) - \hat{S}(T_i)$.
- ▶ The Law of Large Numbers holds for the KM estimator: if $\mathbb{E}(|\phi(Y)|) < \infty$,

$$- \int \phi(y) \hat{S}(dy) \rightarrow \mathbb{E}(\phi(Y)), \quad \text{in probability}$$

The jumps of the KM and IPCW (2/3)

Proposition

Consider a i.i.d. sample (T_i, δ_i) , $1 \leq i \leq n$, of $T = Y \wedge C$ and $\delta = \mathbf{1}_{\{Y \leq C\}}$.

- ▶ Assume that Y and C are independent and there are no ties in the durations T_i .
- ▶ Let $\hat{S}(\cdot)$ (resp. $\hat{G}(\cdot)$) denote the Kaplan-Meier estimator of the survival functions $S(\cdot)$ (resp. $\bar{G}(\cdot)$) of the duration Y (resp. censoring C).

Then,

$$\hat{S}(\{T_i\}) = \frac{1}{n} \times \frac{\delta_i}{\hat{G}(T_i-)}.$$

In particular the KM integral given by the function $\phi(\cdot)$ could be rewritten

$$\sum_{i=1}^n \phi(T_i) \hat{S}(\{T_i\}) = \frac{1}{n} \sum_{i=1}^n \frac{\delta_i \phi(T_i)}{\hat{G}(T_i-)}$$

Proposition

Let $\phi(\cdot)$ be some function with $\mathbb{E}(|\phi(Y)|) < \infty$. Let C independent of Y and define $T = Y \wedge C$ and $\delta = \mathbf{1}_{\{Y \leq C\}}$. Then

$$\mathbb{E} \left[\frac{\delta}{\overline{G}(T-)} \phi(T) \right] = \mathbb{E} \left[\frac{\mathbf{1}_{\{C \geq Y\}}}{\mathbb{E}(\mathbf{1}_{\{C \geq Y\}} \mid Y)} \phi(Y) \right] = \mathbb{E}[\phi(Y)].$$

- ▶ *Inverse Probability of Censoring Weighting (IPCW)* is a general principle allowing to build (asymptotically) unbiased estimates of expectations involving right-censored data
- ▶ The ideal unbiased estimator of $\mathbb{E}[\phi(Y)]$ is

$$\frac{1}{n} \sum_{i=1}^n \frac{\delta_i \phi(T_i)}{\overline{G}(T_i-)}$$

- ▶ Since $\overline{G}(\cdot)$ is unknown, the feasible estimator of $\mathbb{E}[\phi(Y)]$ is

$$\frac{1}{n} \sum_{i=1}^n \frac{\delta_i \phi(T_i)}{\widehat{\overline{G}}(T_i-)},$$

that is, the Kaplan-Meier integral defined by $\phi(\cdot)$!

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Comparing groups of individuals

- ▶ The problem: testing the null hypothesis of equal survival functions among K independent groups under right-censoring
- ▶ Formally, if S_k is the survival function of the individuals in the group $1 \leq k \leq K$, we test

$$H_0 : S_1(\cdot) = \dots = S_K(\cdot)$$

against $H_1 : \exists 1 \leq k \neq l \leq K$ such that $S_k(\cdot) \neq S_l(\cdot)$

- ▶ Let $t_1, t_2, \dots, t_N$ be the **ordered distinct** values in the sample of durations T , censored and uncensored, observed in all groups
- ▶ Notation
 - ▶ $D_{j,k}$ is the number of events (death, failures) observed at time t_j in the group k
 - ▶ $D_j = D_{j,1} + \dots + D_{j,K}$ is the total number of events at time t_j
 - ▶ $R_{j,k}$ = number of individuals 'at risk' at time t_j in group k
 - ▶ $R_j = R_{j,1} + \dots + R_{j,K}$ is the total number of individuals 'at risk' at time t_j

- ▶ The idea for testing is to compare the estimated hazard rates of each group with the one estimated with all observations
→ under H_0 the differences should be 'small'
- ▶ Formally, using some weights w_j , the test is based on the

$$Z_k = \sum_{j=1}^N w_j R_{j,k} \left(\hat{\lambda}_k(T_j) - \hat{\lambda}(T_j) \right) = \sum_{j=1}^N w_j \left(D_{j,k} - R_{j,k} \frac{D_j}{R_j} \right), \quad 1 \leq k \leq K$$

- ▶ One can show that
 - ▶ in general $Z_1, \dots, Z_K$ are not independent (e.g., $Z_1 + \dots + Z_K = 0$ when $w_j \equiv 1$);
 - ▶ under H_0 , the Z_k 's are $N^{-1/2}$ -asymptotically normal;
 - ▶ the asymptotic variance could be easily estimated.
- ▶ Using the properties of $(Z_1, \dots, Z_K)^\top$, one could derive a test statistic under the form of a quadratic form with a χ_{K-1}^2 limit law under H_0
- ▶ Different tests could be derived with different weights w_j
 - ▶ $w_j \equiv 1$ → log-rank test;
 - ▶ $w_j = R_j$ → Gehan test;
 - ▶ *etc.*

The case of two groups ($K = 2$)

► Let

$$Z_1 = \sum_{j=1}^N w_j \left(D_{j,1} - R_{j,1} \frac{D_j}{R_j} \right)$$

► Let

$$V_{Z_1} = \sum_{j=1}^N w_j^2 \frac{R_{j,1}}{R_j} \left(1 - \frac{R_{j,1}}{R_j} \right) \frac{R_j - D_j}{R_j - 1} D_j = \sum_{j=1}^N w_j^2 \frac{D_j R_{j,1} R_{j,2}}{R_j^2} \times \frac{R_j - D_j}{R_j - 1}$$

Proposition

If none of the two groups is asymptotically negligible compared to the other,

$$\frac{Z_1^2}{V_{Z_1}} \rightsquigarrow \chi_1^2 \quad \text{under } H_0.$$

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Regression type modeling

- ▶ In time-to-event modeling, the aim is to model the law of the observations, that is of the realizations of a duration random variable
- ▶ In many cases, the interest could be the conditional law given some covariates (characteristics of the individuals; predictors)
- ▶ This is the so-called *regression* context
- ▶ Recall that in time-to-event analysis, one would rather look for a full description of the conditional law given the covariates, not only for some summaries such as the conditional mean
- ▶ Parametric and nonparametric duration models have been discussed → they could be extended to conditional versions
- ▶ In addition, we will study some semiparametric models with covariates

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Exponential regression

- ▶ A natural way to account for covariates in parametric duration models is to let the parameters depend in the covariates
- ▶ Recall that in a parametric duration model, one considers a family of laws $\mathcal{M} = \{f_\theta : \theta \in \Theta\}$
- ▶ We have introduced two types of such models
- ▶ Exponential:
 - ▶ survivor function $S(y; \lambda) = \exp(-\lambda y)$, $y \geq 0$, with $\lambda > 0$
 - ▶ here $\theta = \lambda$
 - ▶ some authors use a different parametrization $\theta = 1/\lambda$
- ▶ Weibull:
 - ▶ survivor function $S(y; \lambda, k) = \exp(-(\lambda y)^k)$, $y \geq 0$, with $\lambda, k > 0$
 - ▶ here $\theta = (\lambda, k)^\top$
 - ▶ some authors use a different parametrization $\theta = (\lambda^k, k)^\top$

Exponential regression (1/3)

- ▶ The data include realizations of a vector of covariates $\mathbf{Z} \in \mathbb{R}^p$, $p \geq 2$
- ▶ We will suppose the first component of $\mathbf{Z}$ equal to 1
- ▶ If one wants to let λ depending on the covariates, one should consider a function of covariates which takes positive values only!
- ▶ Let define

$$\lambda(\mathbf{Z}; \beta) = \exp(\mathbf{Z}^\top \beta)$$

where $\beta \in \mathbb{R}^p$ is a vector of unknown parameters; the first component of β is the so-called *intercept*

- ▶ This type of modeling could be used with any type of censored/truncated/interval-censored/etc. mechanism
- ▶ Here, we will only consider the case of right-censored durations

Exponential regression (2/3)

- ▶ In exponential regression, one assumes that the conditional law of the duration of interest Y given the covariate vector $\mathbf{Z} = \mathbf{z}$ is exponential with parameter $\lambda(\mathbf{z}; \beta)$

$$Y \mid \mathbf{Z} = \mathbf{z} \sim \text{Exp}(\lambda(\mathbf{z}; \beta))$$

- ▶ Under the right censoring, the data are i.i.d.
 $(T_1, \delta_1, \mathbf{Z}_1^\top)^\top, \dots, (T_n, \delta_n, \mathbf{Z}_n^\top)^\top$, copies of $(T, \delta, \mathbf{Z}^\top)^\top$ with $T = Y \wedge C$ and $\delta = \mathbf{1}_{\{Y \leq C\}}$
- ▶ Here we assume that Y and C are independent given $\mathbf{Z}$

$$Y \perp C \mid \mathbf{Z}$$

- ▶ Under the assumption of an non-informative censoring, the likelihood of the exponential regression model is

$$\mathcal{L}_n(\beta) = \prod_{i=1}^n \{ \lambda(\mathbf{Z}_i; \beta) \exp(-\lambda(\mathbf{Z}_i; \beta) T_i) \}^{\delta_i} \{ \exp(-\lambda(\mathbf{Z}_i; \beta) T_i) \}^{1-\delta_i}$$

Exponential regression (3/3)

- ▶ The log-likelihood is

$$\log \mathcal{L}_n(\beta) = \sum_{i=1}^n \{ \delta_i \log(\lambda(\mathbf{Z}_i; \beta)) - \lambda(\mathbf{Z}_i; \beta) T_i \} = \sum_{i=1}^n \{ \delta_i \mathbf{Z}_i^\top \beta - \exp(\mathbf{Z}_i^\top \beta) T_i \}$$

- ▶ The score equations are

$$\sum_{i=1}^n \mathbf{Z}_i \left\{ \delta_i - \exp(\mathbf{Z}_i^\top \beta) T_i \right\} = 0$$

- ▶ The MLE is not explicit but could be shown to be unique, consistent and $\sqrt{n}$ -asymptotically normal
- ▶ In the case where $\mathbf{Z}_i$ contains only the constant, we recover the MLE mentioned previously (see the first part of the lectures)

Weibull regression (1/2)

- ▶ One could build a parametric regression models for durations starting from the more flexible Weibull family of laws
- ▶ For this purpose, the alternative parametrization of the Weibull where $S(y) = \exp(-\gamma y^k)$ is more user friendly
- ▶ The common modeling is to let k independent of the covariates and to define, as in the exponential model

$$\gamma(\mathbf{Z}; \beta) = \exp(\mathbf{Z}^\top \beta)$$

- ▶ Under the assumption of conditional independence and non-informative censoring, the likelihood of the Weibull regression model is

$$\mathcal{L}_n(\beta; k) = \prod_{i=1}^n \left[\left\{ k T_i^{k-1} \gamma(\mathbf{Z}_i; \beta) \exp \left(-\gamma(\mathbf{Z}_i; \beta) T_i^k \right) \right\}^{\delta_i} \times \left\{ \exp \left(-\gamma(\mathbf{Z}_i; \beta) T_i^k \right) \right\}^{1-\delta_i} \right]$$

Weibull regression (2/2)

- ▶ **Exercise:** write the score equations
- ▶ Again, the MLE is not explicit but could be shown to have good statistical properties (consistent and $\sqrt{n}$ -asymptotically normal)
- ▶ Rather complex algorithms are required to compute MLE, which makes this model less popular among practitioners

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Conditional Kaplan-Meier (1/2)

- ▶ The KM estimator was introduced using an intuition based on the estimate of the hazard function
- ▶ In that story, the idea was that such hazard function estimate is a ratio of probabilities: the probability to fail at time t_k divided by the probability to be at risk at time t_k
- ▶ These probabilities have simple estimates: the sample proportions
- ▶ A natural question is: could one use the same idea in the presence of covariates?
- ▶ The answer is YES!
(This could be also easily understood if we remember KM is a kind of plug-in estimator)
- ▶ But in this case one has to deal with **conditional versions of that probabilities** instead!

Conditional Kaplan-Meier (2/2)

- ▶ Conditional probabilities are easy to estimate nonparametrically if the conditioning variable(s) is (are) discrete with few values
 - ▶ split data in groups according to the values of the conditioning variable(s)
 - ▶ compute the sample proportions in each group
- ▶ Used in the context of the Kaplan-Meier estimator, this yields the so-called *stratified Kaplan-Meier estimator*
- ▶ In the case of continuous covariates, one needs more sophisticated nonparametric method (kernel smoothing, splines, *etc.*)
- ▶ Sometimes, the practitioners simply discretize the continuous variables and use the stratified KM estimator
- ▶ Moreover, since KM is nonparametric, they use such conditional versions of KM to build *ad-hoc* diagnostics for parametric models

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Proportional Hazard Regression Model (1/2)

- ▶ A very popular class of regression models is the so-called *proportional hazard (PH) models*
- ▶ Such models are defined through the hazard function of the duration of interest Y :

$$\lambda(y | \mathbf{z}) = \lambda_0(y)h(\mathbf{z}; \beta), \quad y \geq 0$$

where

- ▶ $\mathbf{z}$ is a possible value for the vector of covariates;
 - ▶ β is a vector of parameters
 - ▶ $h(\cdot; \cdot)$ is a given positive function; e.g., $h(\cdot; \cdot) = \exp(\mathbf{z}^\top \beta)$
 - ▶ $\lambda_0(\cdot) \geq 0$ is the so-called *baseline hazard function*
-
- ▶ When $\lambda_0(\cdot)$ belongs to a parametric family, we say that the PH model is *parametric*
 - ▶ When $\lambda_0(\cdot)$ is an unknown nonnegative function, we say that we have a *semiparametric PH model*

(Don't) touch the constant/intercept

→ a revisited statement (2/2)

- In the case of the usual PH model in duration regression with

$$\lambda(y | \mathbf{z}) = \lambda_0(y) \exp(\mathbf{z}^\top \boldsymbol{\beta}), \quad y \geq 0$$

we have to do the following:

1. if $\lambda_0(y)$ belongs to a parametric family, usually we could include a fixed component equal to 1 in the covariate vector $\mathbf{z}$;
2. if $\lambda_0(y)$ is considered completely unknown (we consider a nonparametric modeling for the baseline hazard function), as it is the case in the semiparametric PH model introduced by Cox (1972), see reference below, then the covariate vector $\mathbf{z}$ **should not** include a fixed component, that means **there is no intercept in the model**

Proportional Hazard Regression Model (2/2)

- ▶ For any $\mathbf{z}$, $\lambda(y | \mathbf{z})$ defines a probability distribution: the law of Y given $\mathbf{Z} = \mathbf{z}$
- ▶ The name of this type of models comes from the following property: for any $\mathbf{z}, \mathbf{z}'$,

$$\frac{\lambda(y | \mathbf{z})}{\lambda(y | \mathbf{z}')} = \frac{h(\mathbf{z}; \beta)}{h(\mathbf{z}'; \beta)}, \quad \forall y \geq 0 \quad (\text{such that } \lambda_0(y) > 0)$$

that is the ratio does not vary over time, and it does not depend on $\lambda_0(\cdot)$!

- ▶ There is a lot of criticism of this property! Even when the model is semiparametric
- ▶ Many alternative regression models have been proposed... one is mentioned below
- ▶ **Exercise:** Show that Weibull and Exponential regression models are PH models. What are the corresponding baseline hazard functions?

Cox's Proportional Hazard Regression Model

- ▶ Cox, David R (1972). "Regression Models and Life-Tables". *J. Royal Statist. Soc., Series B*. **34**(2): 187–220.

- ▶ It is a proportional hazard regression model defined by

$$\lambda(y \mid \mathbf{z}) = \lambda_0(y) \exp(\mathbf{z}^\top \boldsymbol{\beta}), \quad y \geq 0$$

with unknown $\lambda_0(\cdot)$

- ▶ The function $\lambda_0(\cdot)$ is a ‘*nuisance*’ parameter
- ▶ It could be interpreted as the hazard function of a reference individual with zero covariate vector (this because in this model there is no intercept, that means there is no fixed component in $\mathbf{z}$)
- ▶ It is possible to estimate $\boldsymbol{\beta}$ without knowing $\lambda_0(\cdot)$

Full and Partial Likelihood for Cox's PH Model (1/3)

- ▶ In Cox's model, if $S_0(y) = \exp\left(-\int_0^y \lambda_0(t)dt\right)$, we have

$$\Lambda(y | \mathbf{z}) = \exp(\mathbf{z}^\top \beta) \int_0^y \lambda_0(t)dt, \quad S(y | \mathbf{z}) = S_0(y)^{\exp(\mathbf{z}^\top \beta)}$$

and

$$f(y | \mathbf{z}) = \lambda_0(y) \exp(\mathbf{z}^\top \beta) S_0(y)^{\exp(\mathbf{z}^\top \beta)}$$

- ▶ The contributions to the likelihood (under the assumption of conditional independence of Y and C given $\mathbf{Z}$ and non-informative censoring)
 - ▶ uncensored observations $T_i, \mathbf{Z}_i$

$$\left[\lambda_0(T_i) \exp(\mathbf{Z}_i^\top \beta) S_0(T_i)^{\exp(\mathbf{Z}_i^\top \beta)} \right]^{\delta_i}$$

- ▶ censored observations $T_i, \mathbf{Z}_i$

$$\left[S_0(T_i)^{\exp(\mathbf{Z}_i^\top \beta)} \right]^{1-\delta_i}$$

Full and Partial Likelihood for Cox's PH Model (2/3)

- ▶ For simplicity, we only consider the case with no ties in the durations T_i

- ▶ Gathering things, the full likelihood in Cox model is

$$\mathcal{L}_n(\beta) = \prod_{i=1}^n \left[\lambda_0(T_i) \exp(\mathbf{z}_i^\top \beta) \right]^{\delta_i} \left[S_0(T_i)^{\exp(\mathbf{z}_i^\top \beta)} \right]$$

- ▶ Without knowledge of the $\lambda_0(T_i)$'s, it is not possible to maximize directly the full likelihood
- ▶ One possible idea is *profiling out* the nuisance parameter, that is
 - ▶ fix β and search the values $\lambda_0(T_i)$ maximizing the likelihood
 - ▶ plug-in the solution for the $\lambda_0(T_i)$'s to get a likelihood depending only on β
- ▶ The criterion obtained by profiling the full likelihood is the so-called the *Cox's partial likelihood*; its expression is given below

Full and Partial Likelihood for Cox's PH Model (3/3)

- ▶ Let $R(T_i)$ be the set of individuals at risk at time T_i (individuals for which the event didn't happen before T_i)
- ▶ Cox's partial likelihood is

$$\mathcal{L}_{n,Cox}(\beta) = \prod_{i=1}^n \left[\frac{\exp(\mathbf{z}_i^\top \beta)}{\sum_{j \in R(T_i)} \exp(\mathbf{z}_j^\top \beta)} \right]^{\delta_i}$$

- ▶ The maximum $\hat{\beta}$ of the Cox's partial likelihood $\mathcal{L}_{n,Cox}(\beta)$ is consistent and $\sqrt{n}$ -asymptotically normal; it is even efficient in the semiparametric sense.
- ▶ There are many extensions of this basic Cox PH regression model, as for instance to the case of time-varying covariates → beyond our scope!
- ▶ Given the estimate $\hat{\beta}$ of β , one can calculate the estimates of the hazard functions

Breslow's estimator

- ▶ The so-called *Breslow's estimator* of $\Lambda_0(y)$:

$$\hat{\Lambda}_0(y) = \sum_{\{i: T_i \leq y\}} \frac{\delta_i}{\sum_{\{k: T_k \geq T_i\}} \exp(\mathbf{z}_k^\top \hat{\boldsymbol{\beta}})}$$

- ▶ Breslow's estimator of $\Lambda_0(y)$ is an increasing step function (piecewise constant), which has jumps at the uncensored observations
- ▶ The estimator of the conditional cumulative hazard function is

$$\hat{\Lambda}(y | \mathbf{z}) = \hat{\Lambda}_0(y) \exp(\mathbf{z}^\top \hat{\boldsymbol{\beta}})$$

- ▶ Finally, the estimator of the conditional survival function of Y given $\mathbf{Z} = \mathbf{z}$ is :

$$\hat{S}(y | \mathbf{z}) = \exp(-\hat{\Lambda}_0(y) \exp(\mathbf{z}^\top \hat{\boldsymbol{\beta}})). \quad (6)$$

Accelerated Failure Time (AFT) Models (1/3)

- ▶ A competitor of the Cox's PH model is the so-called *Accelerated Failure Time (AFT) Model*
- ▶ In the AFT models, one supposes

$$\log(Y) = \mathbf{Z}^\top \boldsymbol{\beta} + U$$

where U is an unobserved error term independent of $\mathbf{Z}$ (here is not necessarily supposed of zero mean)

- ▶ In the parametric AFT models one supposes that the law of U is given; common examples are
 - ▶ Gumbel, log-normal, log-logistic, generalized gamma, *etc*
- ▶ When the law of U is given, the AFT model parameter $\boldsymbol{\beta}$ could be estimated by ML
- ▶ In semiparametric extension, the law of U is not given (but one has to introduce some additional identification constraints)

Accelerated Failure Time (AFT) Models (2/3)

- ▶ One could rewrite the equation of the AFT model under the form

$$Y = Y_0 \exp(\mathbf{Z}^\top \beta) \quad \text{with} \quad Y_0 = \exp(U)$$

- ▶ Then we have

$$S(y \mid \mathbf{Z}) = \mathbb{P}(Y > y \mid \mathbf{Z}) = \mathbb{P}(Y_0 > y \exp(\mathbf{Z}^\top \alpha) \mid \mathbf{Z}) = S_0(y \exp(\mathbf{Z}^\top \alpha))$$

with $\alpha = -\beta$

- ▶ The terminology AFT becomes more clear: *whereas in a proportional hazards (PH) model, the covariates act multiplicatively on the hazard, in an AFT model the covariates act multiplicatively on time*

Accelerated Failure Time (AFT) Models (3/3)

- ▶ In general, the PH and AFT assumptions cannot be reconciled, and therefore, a model is typically either an AFT model or a PH model, but not both
- ▶ The Weibull regression model (along with its special cases) is the unique model that satisfies both the PH and AFT assumptions